

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA
Screening Test - Kaprekar Contest

(NMTTC- at **SUB-JUNIOR LEVEL** VII & VIII Standards)

Saturday, 31st August 2013.



Note:

- 1) Fill in the response sheet with your Name, Class, the institution through which you appear in the specified places.
- 2) Diagrams are only visual aids; they are not drawn to scale.
- 3) You are free to do rough work on separate sheets.
- 4) Duration of the test: 2 p.m. to 4. p.m. - 2 hours.

PART – A

Note:

- Only one of the choices A, B, C, D is correct for each question. Shade that alphabet of your choice in the response sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).
- For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{2}$ mark.

1. If $a + b + c = 0$ where a, b, c are non zero real numbers, then the value of $(a^2 - bc)^2 (b^2 - ca)^2 (c^2 - ab)^2$ is

(A) 1 (B) abc (C) $a^3 + b^3 + c^3$ (D) 0

2. In the adjoining incomplete magic square the sum of all numbers in any row or column or diagonals is a constant value. The value of x is

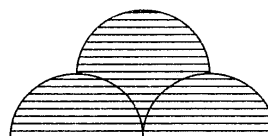
17				
23	5			
	6	12	20	x
10			21	
11				9

(A) 18 (B) 24 (C) 22 (D) 16

3. Aruna, Bhanu and Rita have some amount of money. The ratio of the money of Aruna to that of Bhanu is 7:15 and the ratio of the money of Bhanu and Rita is 7:16. If Aruna has Rs. 490, the amount of money Rita has is (in Rupees)

(A) 1500 (B) 1600 (C) 2400 (D) 3600

4. In the adjoining figure, all the three semicircles have equal radius of 1 unit. The area of the shaded portion is



(A) $\pi + 2$ (B) 5 (C) $\frac{3}{2} + 1$ (D) 4

5. The sum of three different prime numbers is 40. What is the difference between the two biggest ones among them?
- (A) 8 (B) 12 (C) 20 (D) 24
6. Peter has written down four natural numbers. If he chooses three of his numbers at a time and adds up each triple, he obtains totals of 186, 206, 215 and 194. The largest number Peter has written is
- (A) 93 (B) 103 (C) 81 (D) 73
7. There are four non-zero numbers x, y, z and u . If $x = yz$, $y = zu$, $z = ux$, then the value of $\frac{x}{y} + \frac{y}{z} + \frac{z}{u} + \frac{u}{x}$ is equal to
- (A) 1 (B) $\frac{1}{2}$ (C) 0 (D) $\frac{1}{2}$
8. The natural numbers from 1 to 20 are listed below in such a way that the sum of each adjacent pair is a prime number.
20, A, 16, 15, 4, B, 12, C, 10, 7, 6 D, 2, 17, 14, 9, 8, 5, 18, E. The number D is
- (A) 1 (B) 3 (C) 11 (D) 13
9. How many ordered pairs of natural numbers (a, b) satisfy $a + 2b = 100$
- (A) 33 (B) 49 (C) 50 (D) 99
10. The value of $(\sqrt{5} \ \sqrt{6} \ \sqrt{7}) (\sqrt{5} \ \sqrt{6} \ \sqrt{7}) (\sqrt{5} \ \sqrt{6} + \sqrt{7}) (\sqrt{6} + \sqrt{7} \ \sqrt{5})$ is
- (A) $3\sqrt{210}$ (B) 210 (C) $4\sqrt{210}$ (D) 104
11. The least number which, when divided by 52 leaves a remainder 33, when divided by 78 leaves 59 as remainder and when divided by 117 leaves 98 as remainder is
- (A) 553 (B) 293 (C) 468 (D) 449
12. A sum of money is divided between two persons in the ratio 3:5. If the share of one person is Rs. 2000 more than that of the other, then the sum of money is (in rupees)
- (A) 6000 (B) 8000 (C) 10,000 (D) 12,000
13. Two numbers are respectively 20% and 50% more than a third number. What percent the first number is with respect to the second number?
- (A) 70% (B) 30% (C) 80% (D) 60%

14. There are some toys. One third of them are sold at a profit of 15%, one fourth of the total are sold at a profit of 20% and the rest for 24% profit. The total profit is Rs. 3200. The total price of the toys is (in rupees)

(A) 32000 (B) 64000 (C) 16000 (D) 48000

15. The radius of a circle is increased by 4 units and the ratio of the areas of the original and the increased circle is 4:9. The radius of the original circle is

(A) 6 (B) 4 (C) 12 (D) 8

PART – B

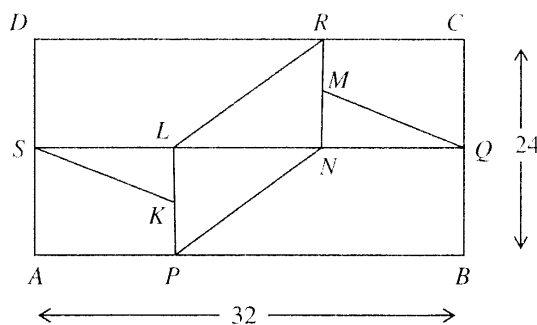
Note:

- Write the correct answer in the space provided in the response sheet.
- For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{4}$ mark.

16. A natural number less than 100 has remainder 2 when divided by 3, remainder 3 when divided by 4 and remainder 4 when divided by 5. The remainder when the number is divided by 7 is _____.

17. The units digit of $(2013)^{2013}$ is _____.

18. In the adjoining figure $ABCD$ is a rectangle with sides 24 units, 32 units. SKL , PLN , QNM , RNL are congruent right angled triangles. The area of each triangle is _____.



19. The number of natural numbers 'n' for which $\frac{(n+2)(n+2)}{(n-3)}$ is a natural number is _____.

20. The product of two natural numbers 'a' and 'b' divides 48. a, b are not relatively prime to each other. The number of pairs (a, b) where $1 < a, b < 48$ ($a \neq b$) is _____.

21. The value of the expression $\left(\frac{a^{-\frac{1}{2}} a^{\frac{1}{2}}}{a^{\frac{1}{2}} a^{-\frac{1}{2}}} - \frac{1 - a^{\frac{1}{2}}}{\sqrt{a} + a^{-\frac{1}{2}}} \cdot a^{\frac{1}{2}} \right)$ is $\frac{2}{K^3}$ where $a = (2013)^2$. The value of K is _____.

22. AB and AC are two straight line segments enclosing an angle 70° . Squares $ABDE$ and $ACFG$ are drawn outside the angle BAC . The diagonal FA is produced to meet the diagonal EB in H . then $\angle EAH =$ _____.
23. In the pentagon $ABCDE$, $D = 2B$ and the angles A, C, E are each equal to half the sum of the angles B and D . The maximum interior angle of the pentagon is _____.
24. The ratio of a two digit number and the sum of its digits is 4:1. If the digit in the units place is 3 more than the digit in the tens place, then the number is _____.
25. There is a famine in a place. But there is sufficient food for 400 people for 31 days. After 28 days 280 of them left the place. Assuming that all of them consume the same amount of food, the number of days for which the rest of the food would last for the remaining people is _____.
26. A triangle whose sides are integers has a perimeter 8. The area of the triangle is _____.
27. If $(10^{2013} + 25)^2 - (10^{2013} - 25)^2 = (\sqrt{10})^n$ then the value of n is _____.
28. ABC and ADC are isosceles triangles with $AB=AC=AD$. $\angle BAC = 40^\circ$, $\angle CAD = 70^\circ$. The value of $\angle BCD + \angle BDC =$ _____.
29. There are three persons Samrud, Saket and Vishwa. Samrud is twice the age of Saket and Saket is twice the age of Vishwa. Their total ages will be trebled in 28 years. The present age of Samrud is _____.
30. The number of three digit numbers of the form $ab5$ which are divisible by 9 is _____.